

DP

SELECTED RESEARCH / SECTOR REPORT

DIMITRIS PECHLIVANIDIS

RWA Perpetuals

Where around-the-clock equity and commodity trading
finds demand

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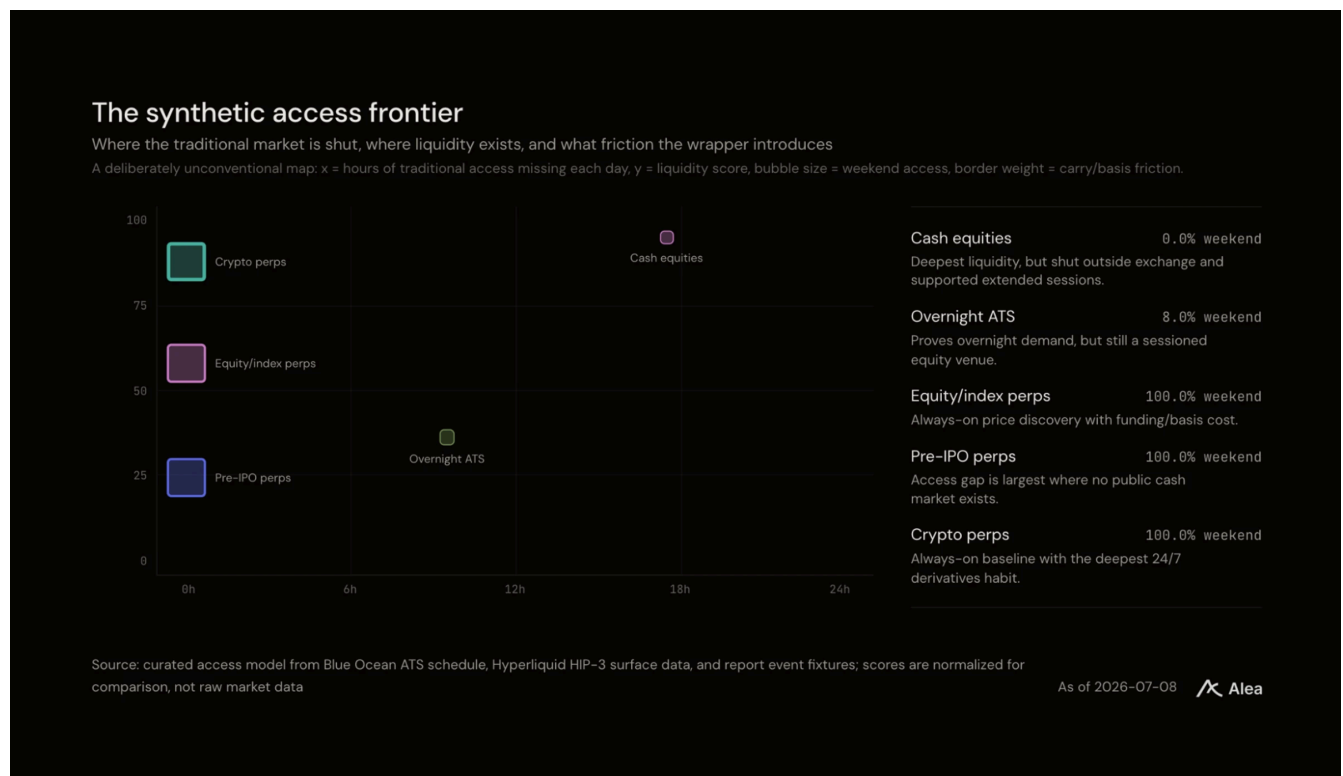
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Executive Summary

Tokenized RWA perpetuals are gaining traction as open interest across both DEX and CEX venues hit \$4.3B, a 15x gain year-to-date.

Their appeal lies in addressing two limitations of traditional financial systems. They let you price macro risk even when markets are closed on weekends, and they give retail a way into private companies that was never on the table before.



U.S. equity trading volumes now cross \$1T/day, up >30% year-over-year. However, this liquidity is **limited to its 930AM to 4PM ET schedule**. Blue Ocean ATS, the largest dedicated overnight venue, averaged roughly \$1B a night in 2025, about **0.1% of daytime volume**, drawing mostly APAC and European retail trading U.S. stocks in their own daylight hours.

The night shift: geography of the overnight bid

The US overnight window against the local daytime of the world's financial centers

When the US cash market is shut (20:00–04:00 ET), the entire Asia-Pacific business day is live — the overnight tape is their daytime. Europe only wakes for the final hour or two; the Americas are in their own night. Timezone math, with the one cited cohort figure: Korea alone is ~35% of Blue Ocean's global volume.



Asia-Pacific coverage

100%

of the overnight window

Europe coverage

25%

the final two hours only

Korea

35%

of Blue Ocean global volume, from 65%
two years ago

Blue Ocean ADV

\$3.5B

130M shares a night, 30+ countries

■ Asia-Pacific daytime ■ Europe daytime ■ In local night

Window: 20:00–04:00 ET, cash market closed · local daytime 08:00–20:00 · boreal-summer offsets (±1h in winter)

Source: Alea Analytics timezone model of the 20:00–04:00 ET overnight window against local daytime (boreal-summer offsets; ±1h in winter); cohort share from Blue Ocean CEO Brian Hyndman, Aju Business Daily, 20 Mar 2026. Captured 2026-07-08

As of 2026-07-08

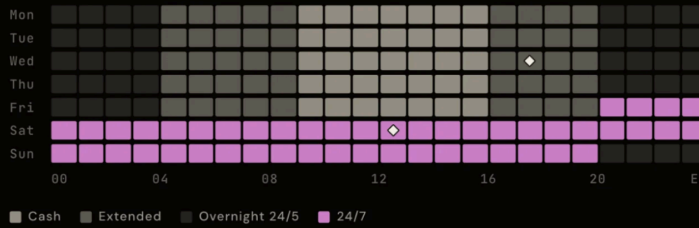
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The 28th Feb (Saturday) U.S. strike on Iran put 24/7 trading in the spotlight. Every regulated venue was **closed on the weekend**, and Hyperliquid was the **only place** where traders could price the shock. Worth noting the flow clustered in crude, one commodity. Even so, it signals weekend appetite for macro pricing more generally. Blue Ocean's next session on Sunday evening saw a **4x spike in volume** compared to what it was earlier that week. CME, which opens on Sunday 6PM ET, saw Oil contracts nearly **doubling** the prior week's average.

The 168-hour week, and who is open for it

Every ET clock-hour coloured by the deepest venue trading, and the weekend only crypto can price

Regulated coverage steps up from the cash session (09:30–16:00) to extended ECN hours to the Blue Ocean 24/5 / announced Nasdaq 23/5 ceiling, but stops at Friday evening. The 48 weekend hours in accent are coverable only on 24/7 crypto rails, the moat that survives even after regulated venues run 23/5. Diamonds mark where the report's hours-driven events landed.



Cash session (RTH) 32.5h

Extended hours (ECN) 80h

Overnight ATS + announced 23/5 120h

Crypto / RWA perps (24/7) 168h

48h

of every week, the weekend, 28.6% of the clock, can be priced only on 24/7 crypto rails. This is the moat that survives after regulated venues run 23/5.

◆ Where the report's events landed: WTI crude (Sat, weekend moat) · MU (Wed, after-close).

Source: public venue schedules, NYSE/Nasdaq RTH 09:30–16:00 ET, pre/post-market ECN 04:00–20:00, Blue Ocean ATS 24/5 + announced Nasdaq 23/5 (Sun 20:00 → Fri 20:00), crypto/RWA perps 24/7. Deterministic coverage model, not market data · Event placements use each event's real date at its band-representative hour

As of 2026-07-10 Alea

The other big pull is private markets. As companies nowadays **stay private longer**, most pre-IPO value accrues to **insiders** and **accredited funds**, leaving retail unable to reach names like Anthropic and OpenAI. Pre-IPO perps make these assets tradeable.

How each class behaves, not just how big it is

Turnover, carry, concentration and realized move by asset class

Turnover is 24h volume + open interest; funding is median annualized APR with the inter-quartile carry spread; concentration is the OI Herfindahl index (1 = one market holds all the OI) with the dominant ticker; move is the median absolute 24h price change. Live across every HIP-3 builder DEX.

Asset class	Markets	24h volume	Open interest	Turnover	Funding APR	OI concentration	Median move
Equities	88	\$240M	\$2.1B	0.1x	+0.0% IQR 33.9pp	0.09 SKHX 23.7%	0.5%
Commodities	23	\$203M	\$642M	0.3x	+0.0% IQR 2.3pp	0.26 CL 34.8%	0.0%
Indices	25	\$78M	\$918M	0.1x	+0.0% IQR 0.0pp	0.51 SP500 59.2%	0.0%
FX	6	\$461K	\$31M	0.0x	+0.0% IQR 0.0pp	0.51 JPY 58.4%	0.0%
Pre-IPO	4	\$157K	\$1M	0.1x	+0.0% IQR 1.4pp	1.00 QNT 100.0%	0.3%

Source: Hyperliquid API, HIP-3 builder-deployed perp markets; per-class turnover, annualized funding, OI Herfindahl concentration and median 24h move

As of 2026-07-12 Alea

Both weekend macro pricing and private-market demand now show up as RWA perpetuals. The rest of this report is about what happens next, and who's positioned to own this flow and capture its value.

Key Takeaways

- **RWA perps solve time and access, not tokenization** - Demand comes from pricing risk while markets are closed and trading names no exchange can list.
- **Overnight demand was always present** - Blue Ocean cleared ~\$1B a night in 2025, led by APAC and European retail. Korea alone is 35% of recent volume.
- **Weekends are crypto's uncontested window** - Regulated venues trade 5 days a week, and even extended overnight sessions go dark over the weekend. When the U.S. struck Iran on a Saturday, only 24/7 venues like Hyperliquid could price the shock.
- **Pre-IPO perps are retail's only route into private markets** - Companies stay private ~14 years with ~41% of recent IPO returns accrued pre-listing. Perps price these names continuously, allowing retail to get exposure without needing accreditation.
- **DEXs win on breadth, CEXs on distribution** - Permissionless listing captures long-tail tickers and jurisdictions while CEXs capture KYC'd flow from emerging-market retail and mandated institutions.

Overnight Equity Trading Demand

Global trading interest has concentrated in a handful of **U.S.-listed** AI names. The “[AI Big 10](#)”, comprising MAG7 plus Broadcom, AMD, and Micron, accounts for a combined market value of roughly \$26T, about 38% of the [S&P500](#), and 35% of the [total US stock market](#). This level of concentration [places AI alongside](#) the peaks of the 1970s Nifty Fifty and the 2000s dot-com era.

That liquidity sits **almost entirely on U.S. venues**, leaving retail and institutional traders worldwide to reach it through local brokerages. Platforms like Interactive Brokers send overnight orders to [Blue Ocean ATS](#), an SEC-registered alternative trading system (ATS) that matches orders in U.S. stocks while the exchanges are closed.

Price discovery happens primarily within the 9:30AM to 4:00PM ET window. That's when the exchanges, their market makers, and the opening/closing auctions all operate. During after-hours, exchanges close and trading continues only on [Electronic Communication Networks \(ECNs\)](#). Extended-hours sessions carry about [11% of U.S. equity volume](#), with bulk of the volume happening in the hours right before and after regular hours trading.

The news does not keep those hours. Companies deliberately schedule earnings and sensitive news for after-hours, and macro prints land pre-market. Weekends are the extreme case, with every venue closed and nothing to absorb a shock until Sunday night.

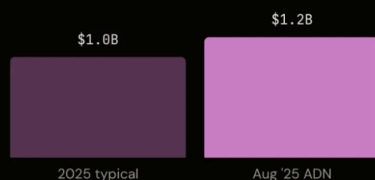
Blue Ocean ATS was built to serve that unmet demand. It runs the largest dedicated overnight session for U.S. equities, matching orders from 8:00PM to 4:00AM ET, Sunday night through Friday morning, and it [cleared over \\$370B](#) in overnight notional in 2025, roughly \$1B a night.

Overnight equity demand is compounding

Blue Ocean ATS published average-night and record-session notional, with no inferred annual total

Published average per-night notional through 2025. Blue Ocean has no public time series, so these are cited milestone figures, not a continuous 7-day average.

>+20% August ADN vs the published ~\$1B typical night



Typical night
~\$1.0B

August ADN
>\$1.2B

August uplift
>+20%

Record session
\$4.9B

Peak: \$4.9B on 2025-04-07, All-time notional record as reported on 25 Sep 2025.

Source: Blue Ocean Technologies, 25 Sep 2025; October 2025 Blue Ocean Newsletter. Published snapshots, not a continuous tape

As of 2025-11-05 Alea

That nightly figure is about 0.1% of daytime volume, which reads small until you trace where it came from: nightly share volume climbed from under a million shares in 2022 to 30 to 50 million by 2025. **Retail drove that climb**, concentrated in **APAC and Europe** trading US names during their own working hours. The demand always existed, just in timezones the US session never reached. And it keeps growing.

Blue Ocean demand, not just a headline

Published overnight notional and disclosed share-volume records, with the 2025 average as the baseline

The report asks for trades and notional. Blue Ocean does not expose a public tape; public sources do not disclose trade count, so this panel separates notional milestones from disclosed share volume.

Typical night
~\$1.0B

August ADN floor
>\$1.2B

August vs typical
>+20%

Disclosed record
159.5M shares



Oct 13 104.8M shares
October high: 104.797M shares and \$6.094B notional

Nov 4 116.9M shares
Korean retail reconnection: 116.9M shares and \$4.106B

Nov 5 159.5M shares
Korean retail reconnection: 159.5M shares and \$5.69B

Source: Blue Ocean Technologies, 25 Sep 2025; October 2025 Blue Ocean Newsletter. Published snapshots, not a continuous tape

As of 2025-11-05 Alea

Korea accounts for 35% of Blue Ocean's recent volumes, mostly retail investors. Thanks to KOSPI's historic AI-fuelled surge, Korean retail flocked to both the local and US stock markets, accounting for 60-70% of trading volumes locally.

Despite the surging retail interest globally, the platforms supporting overnight trading are still closed on weekends. Blue Ocean runs 24/5, Nasdaq recently announced support for 23/5 trading this year, but this still means the market closes on Friday and reopens Sunday night.

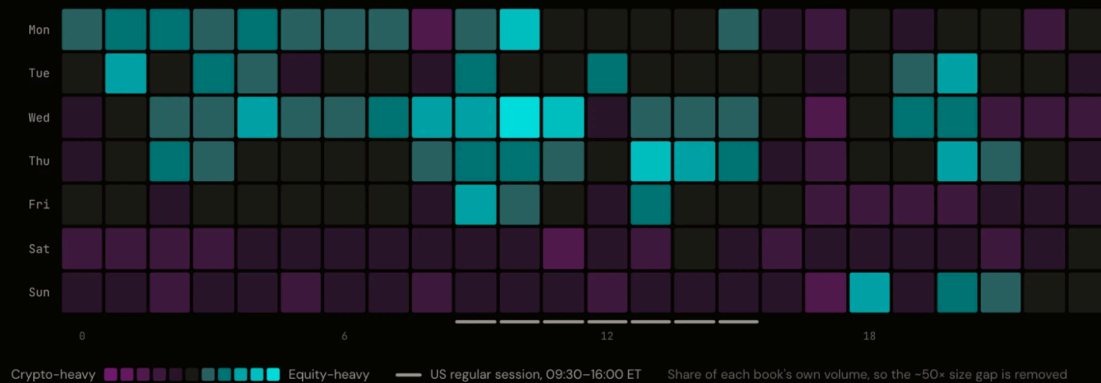
Crypto's Already 24/7 Rails

Crypto is built exactly for meeting this demand. Blockchains settle continuously, unlike traditional clearing houses that only work on business hours. RWA perps built on crypto rails trade straight through the weekend, exactly when regulated venues go dark.

Where equity trades unlike crypto

Equity minus crypto share of each ET hour, the divergence in one grid

Warm cells are where equity perps over-concentrate versus the near-uniform crypto book (the US cash session); cool cells are where they thin out, overnight and on weekends. Shares of each book's own volume, so the size gap is removed.



Source: Hyperliquid API, hourly candle notional, last 8 weeks, equity minus crypto share per ET hour

As of 2026-07-12 Alea

Perps work using funding rates and pulling the price of the underlying asset. This offers price exposure **without having to hold** the underlying asset or shares, making them highly capital efficient for speculation and hedging. TradeXYZ introduced the first RWA perps using this model on Hyperliquid's HIP-3, listing U.S. Korean and even Chinese stocks, as well as a licensed S&P500 contract that settles in USDC.

From a standing start to a \$3B book

HIP-3 network open interest since launch, with the run-rate extended

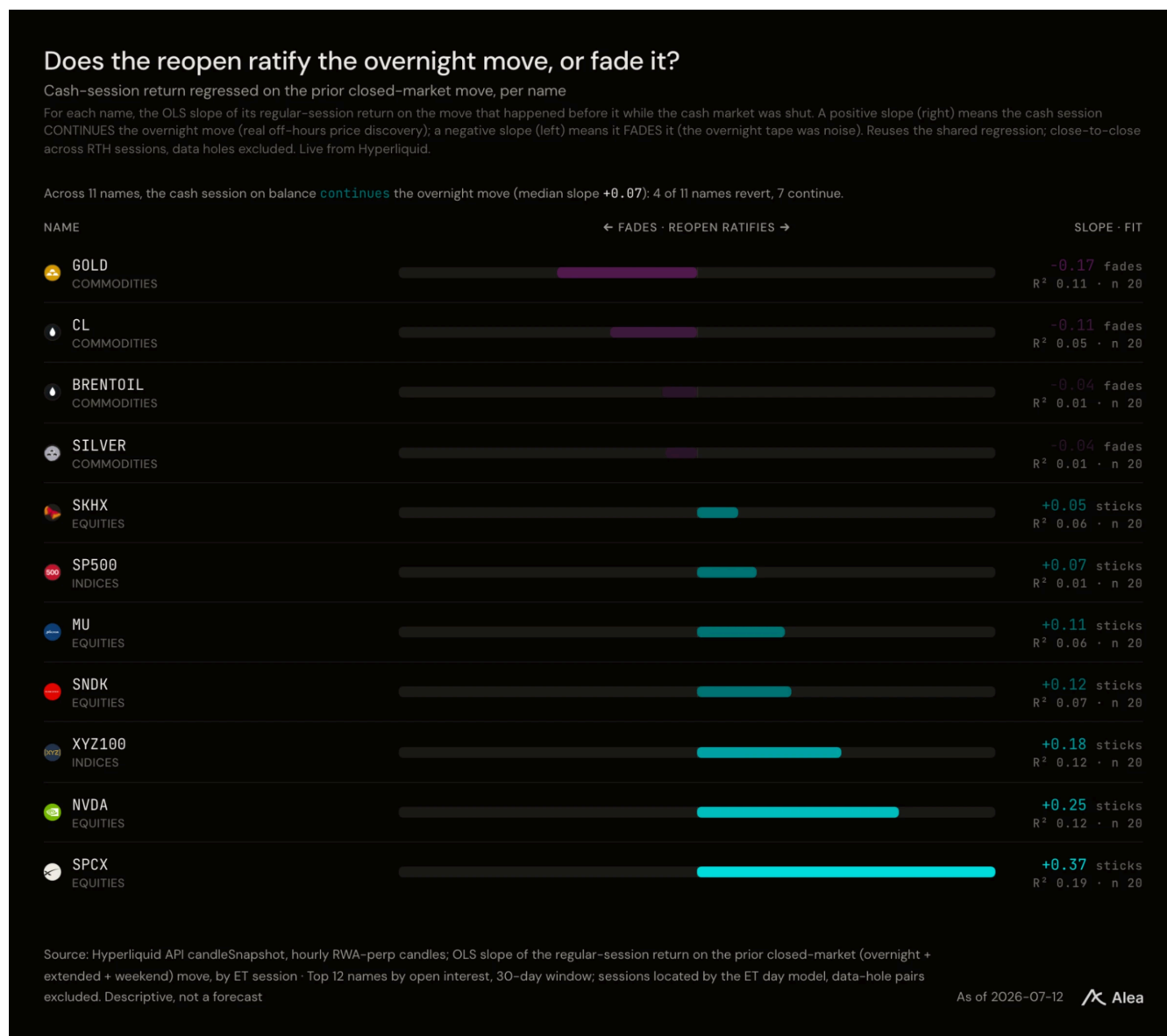
The RWA-perp market went from zero at the 2025-10-13 HIP-3 launch to a live network open interest today. The dashed extension is an illustrative trend at the trailing run-rate — not a forecast. Per-class history is network-only for now and is captured from today forward.



Source: Ioris.tools HIP-3 dashboard (network open interest and 30-day volume) joined to the live Hyperliquid HIP-3 cross-section; trajectory anchored at the 2025-10-13 mainnet launch. Dashed extension is a trailing-run-rate trend, not a forecast; per-class growth is captured going forward

As of 2026-07-12 Alea

During U.S. hours, the perp marks against a [live oracle](#) read of the equity or index price, with funding tugging it back toward spot. When that reference disappears Friday at 4pm, the venue has to let the [mark](#) float on its own order book to price the weekend at all, while capping how far and how fast it can drift so a thin book doesn't gap on one liquidation cascade. TradeXYZ uses [discovery bounds](#) to limit drastic price movements on the weekends.

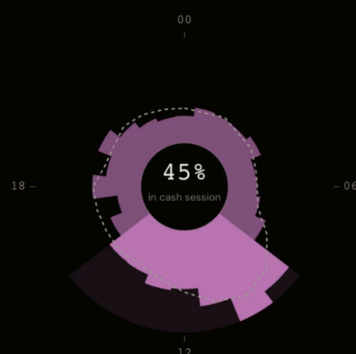


Is there real demand here, or are crypto traders just passing time? To answer that question we look at where RWA perp volume concentrates overnight and during weekends. Crypto-native perps trade at 3pm as readily as 3am, so its volume spreads fairly evenly across the clock and carries little signal. Market opens still carry a large inflow of liquidity, hence the spike in both RWA and crypto perps during cash sessions.

The 24-hour clock

When the equity-perp book trades, by ET hour, against crypto's always-on ring

Share of each day's volume per hour. The equity rose spikes past the crypto circle across the US cash session and falls inside it overnight, equity perps still trade like equities. The gap is the room the migration has left.



Cash-session intensity

1.1×

vs crypto, same hours

Still trades overnight

25%

20:00–04:00 ET

Busiest hour

10:00

9% of the day

Room to 24/7

+16pp

in-session vs a flat book

Equity perps, share of daily volume by hour — Crypto perps, near-uniform 24/7 ring

US cash session, 09:30–16:00 ET

Midnight ET at top, noon at bottom · each book to its own daily total

Source: Hyperliquid API, hourly candle notional over the last 8 weeks, collapsed to ET hour-of-day share

As of 2026-07-12

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RWA perp volume builds through the US cash session, peaking in the 10:00 open hour with **9% of the day's flow** and 1.1x of crypto over the same hours, yet 25% of all volume still prints between 8:00PM and 4:00AM, when no US equity venue is matching orders.

The market-hours gap, measured

Closed-hours and weekend activity: equity/index perps vs crypto perps and overnight equities

Closed hours = everything outside 09:30–16:00 ET: extended 04:00–09:30 and 16:00–20:00, overnight 20:00–04:00, plus weekends. Weekend = Saturday and Sunday ET.

Equity/index perps

Hyperliquid HIP-3

Closed hours

Weekend

CL, BRENT OIL, SKHX, SP500, XYZIOO basket

65.1% closed

11.5% weekend

Crypto perps

Hyperliquid core

Closed hours

Weekend

BTC, ETH, HYPE, SOL, ZEC basket

71.0% closed

19.3% weekend

Overnight equities

Blue Ocean ATS

Closed hours

Weekend

Session schedule: 20:00–04:00 ET, Sunday–Thursday; no public weekend distribution tape.

100.0% closed

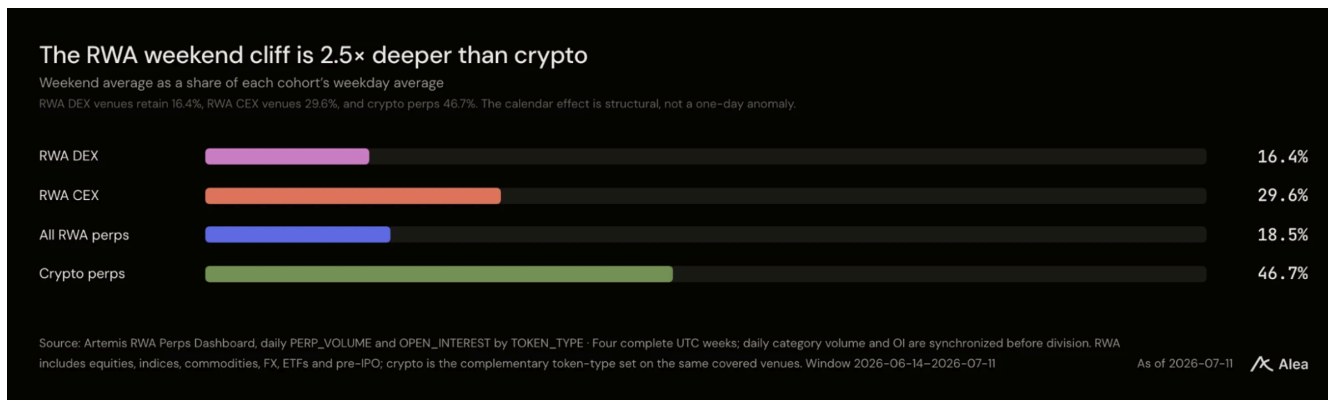
n/a weekend

Source: Hyperliquid API hourly candle notional, last 8 weeks; Blue Ocean ATS published session schedule

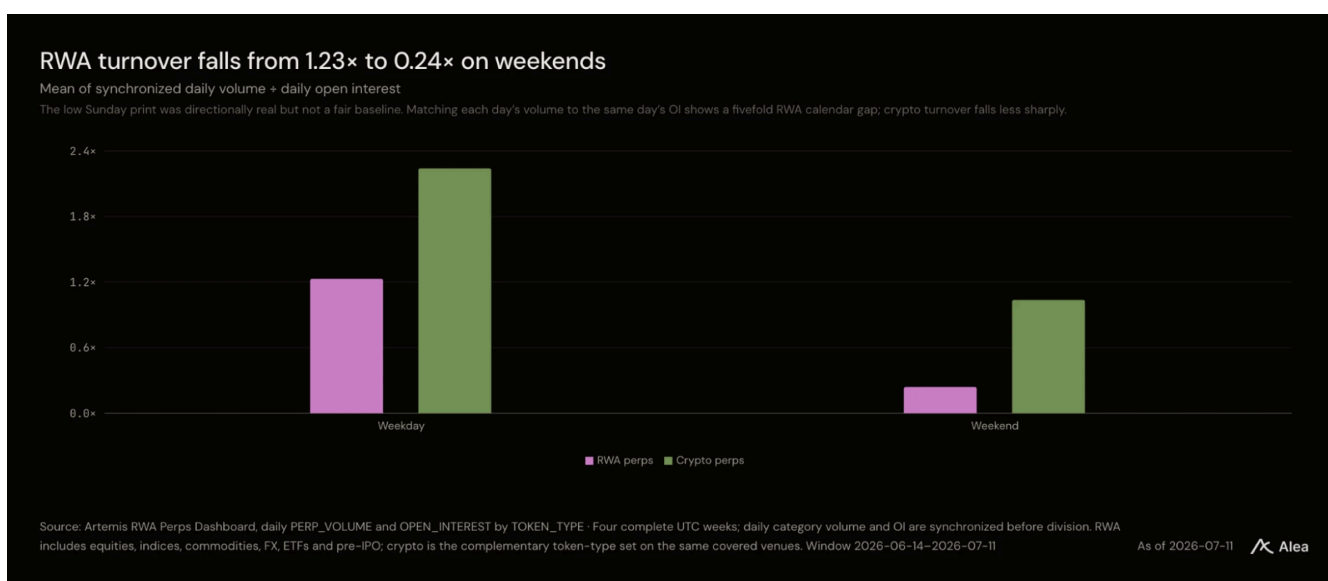
As of 2026-07-12

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Weekend share isolates the demand no regulated venue can contest. Any weekend equity or index flow has nowhere to route except crypto rails. Over the trailing eight weeks of Hyperliquid data, RWA perps printed **65.1%** of their notional outside the 9:30AM to 4:00PM ET cash session and **11.5% on weekends**, when Blue Ocean runs no session at all. Crypto perps ran 71.0% and 19.3% on the same clock, close to their round-the-clock baseline. The RWA perp book still leans toward US hours, which is expected since earnings and macro prints land on that schedule. Across all DEX and CEX platforms, RWA perps retain just 18.5% of its weekday average on weekends, falling from \$4.9B a day to ~\$900M.



Looking at daily notional divided by open interest separates held risk from recycled flow. RWA perps turn over 1.23x on weekdays and 0.24x on weekends while open interest itself barely moves. Volume may see a decline but open interest barely moves as traders carry positions through the weekend.



Pre-IPO Equity Perpetuals

Outside of public equities that are limited by trading hours, crypto rails are also capable of serving demand for markets **limited by access**. Private markets are a [\\$5T market](#) but they gate access behind accredited-investor rules and multi-month lock-ups. They also clear at prices most participants never see.

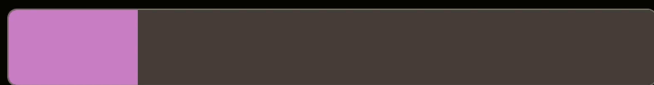
Companies today stay private for far longer than they used to, [14 years today vs <10 years pre-2020](#). The value that is created before an IPO accrues almost entirely to a small closed set of late-stage investors and insiders. Select IPOs from 2014-2019 generated [21% of returns](#) from private markets, but those from the 2020-2023 batch generated almost 2x, ~41% from private markets.

More value forms behind the private wall

Share of market capitalization formed before listing, shown as rounded cohort thresholds from a cited a16z snapshot

CITED A16Z SNAPSHOT · NOT LIVE DATA

2014–2019 IPOs <20% private · >80% public



Recent IPO cohort >50% private · <50% public



0 25 50 75 100

■ Private · pre-IPO, walled off to retail ■ Public · post-IPO, anyone can buy

<20% → >50%

of market capitalization formed before listing in the recent cohort, versus less than one-fifth for the median 2014–19 company.

Scope caveat: the newer companies have had less time to compound after listing.

>30 pts minimum widening implied by the two rounded cohort thresholds

~14 yrs median age at IPO now, versus 5 years two decades ago

\$4.7T private tech unicorn value in the September 2025 article; the January 2026 deck says it crossed \$5T

REPRESENTATIVE PUBLIC COHORT NAMES · LOGOS FOR ORIENTATION, NOT A COMPLETE SAMPLE

Uber



Airbnb



Coinbase



DoorDash



Shopify



Snap



Robinhood



Snowflake



Roblox



Spotify

Still-private examples named by a16z include SpaceX, OpenAI, Anthropic. A pre-IPO perp can provide continuous price exposure, not ownership of private shares; funding, liquidation and venue risk still apply.

Source: a16z, Private Markets Are The New High-Growth Public Markets (Sep. 2, 2025); a16z State of Markets (Jan. 22, 2026). Rounded cohort thresholds and cited valuation snapshots; not live market data. The recent cohort has had less time to compound in public markets. · As of Jan 22, 2026

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A pre-IPO perp is the only exposure that reaches this demand. It settles in USDC, requires no accreditation, and prices a company continuously that otherwise trades a few times a year in private rounds. Funding rounds usually provide a rough floor of what the pre-IPO perp should be worth, but the time between rounds or leading up to the IPO is unpredictable. Theoretically, if there is large unmet demand from retail investors in private companies, pre-IPO perps should also trade at a premium. To investigate this, we look at how Cerebras (CBRS), SpaceX (SPCX) and Quantinuum (QNT) ended up opening on their IPO days relative to what pre-IPO perps were saying.

Pricing Accuracy at the Open

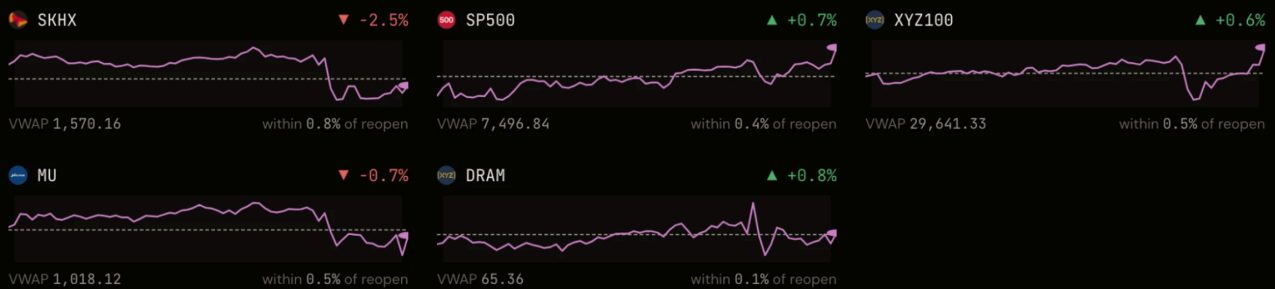
Overnight, weekend, and pre-IPO perp trading exists to price events that traditional market infrastructure historically left unpriceable until it reopened. One way to judge how efficiently they do that is to check the **off-hours price** against the number the primary market clears at **when it reopens**. On regularly traded names, we can see that the closed-window VWAP managed to price the reopen within a **median 0.5%** over the July 4 long weekend.

The last weekend, priced in the dark

Perp price from Fri, Jul 3 close to Mon, Jul 6 reopen

Each spark is one perp's price through the closed weekend (cash shut), ending at the reopen. The net move is what the perp discovered while cash couldn't trade; the VWAP line is the closed-window average, and how close it lands to the reopen is the accuracy. Recomputed live each week.

5 equity perps · median weekend move 0.7% · closed-window VWAP priced the reopen within a median 0.5%



Source: Hyperliquid API, hourly xyz: candles. Perp VWAP (typical × volume) over the Friday-close→Monday-reopen window vs the perp's oracle price at the reopen

As of 2026-07-12 Alea

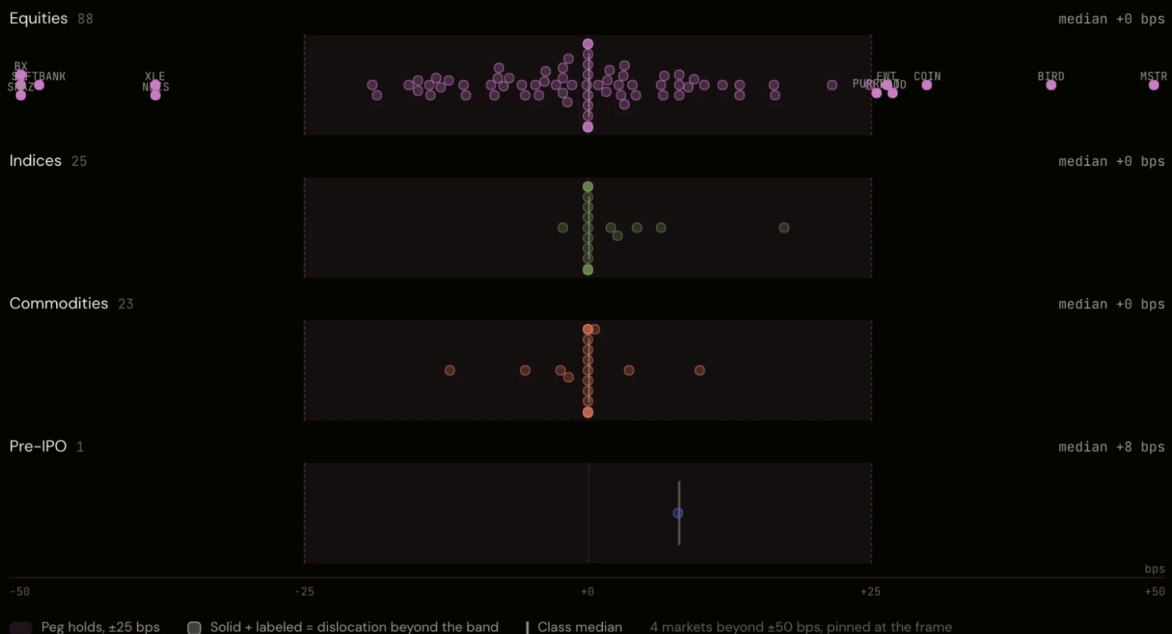
Across 137 RWA perp markets, the mark held within 25bps of its oracle in 125 of them, and the median market sat 3 bps off its reference, tighter than the bid-ask spread on many mid-cap stocks.

Does the peg hold overnight?

Mark-vs-oracle basis for every RWA perp, in basis points, by class

Each dot is one market's premium to its oracle. The shaded band is ±25 bps; markets inside it track their underlying tightly even while the cash market is shut. The few outside are the real dislocations, named.

Peg holds within ±25 bps for 125 of 137 markets (91.2%) · median |basis| 3 bps · 12 dislocations

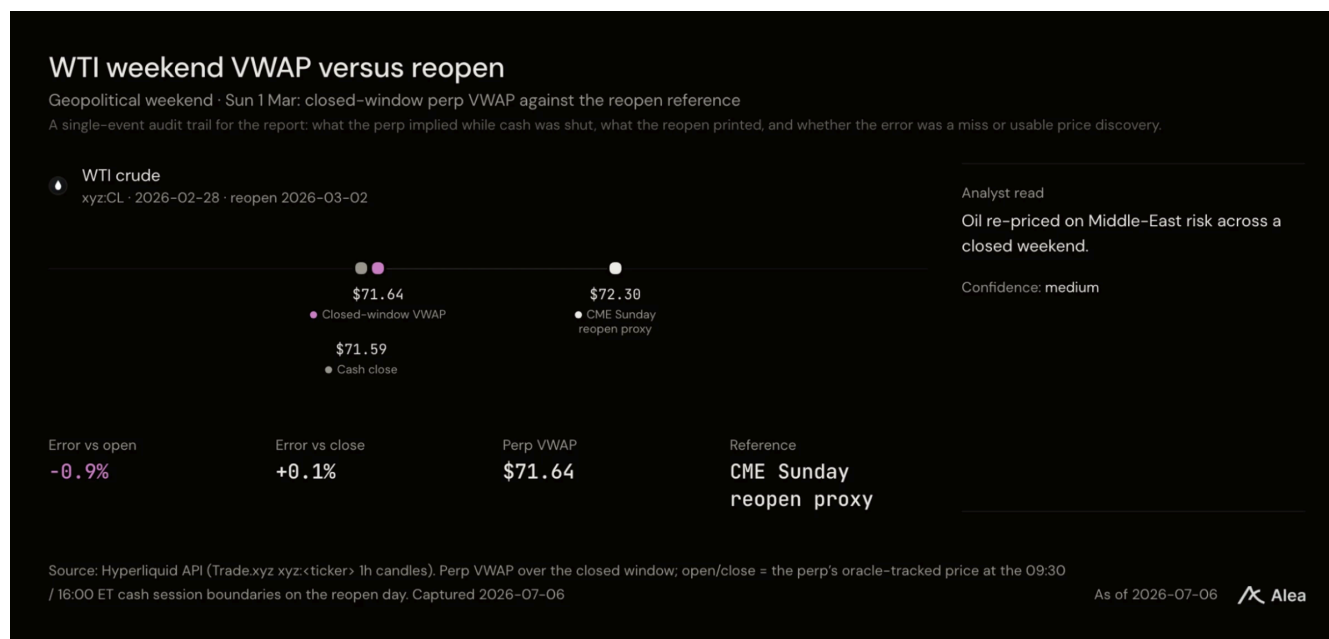


Source: Hyperliquid API, mark-vs-oracle premium across every HIP-3 RWA and real-price pre-IPO market, in basis points

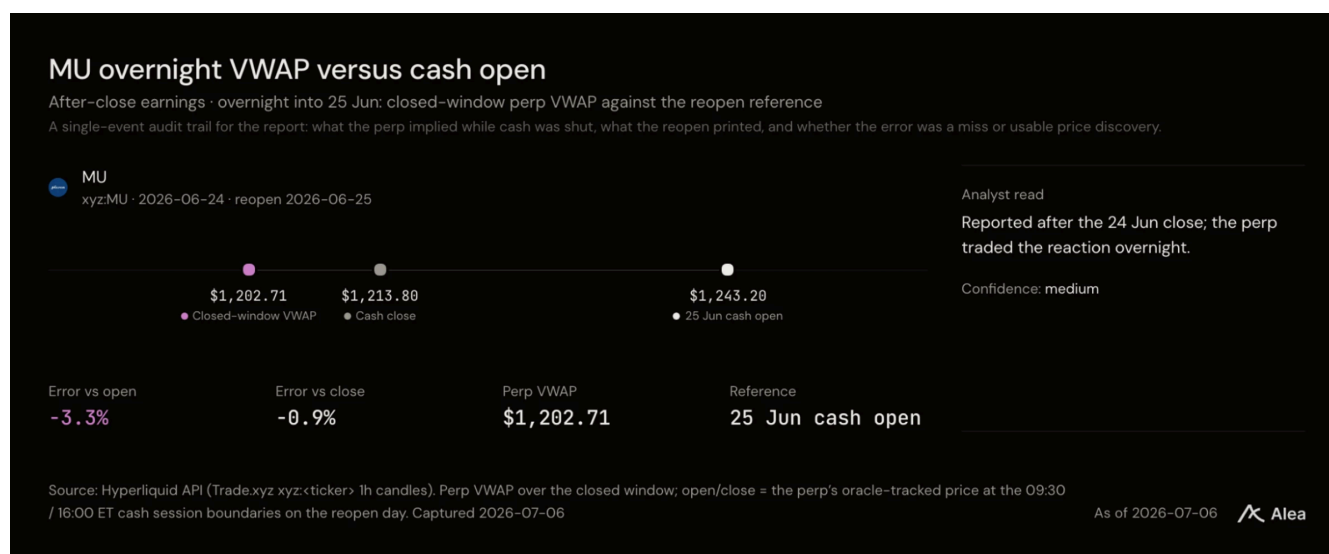
As of 2026-07-12 Alea

If we look at more volatile events such as geopolitical shocks, earnings, or IPO days, the opening ranges might differ. Across every event we test, we use the perp's VWAP over the market close and compare it against the underlying's opening/closing price to see what range it falls within each.

The first event is the Iran conflict which started on Saturday, 28 February, looking at WTI Crude. CME closes Friday and reopens Sunday night, so the HIP-3 perp was the only liquid venue where this could be priced in. WTI crude closed-window VWAP printed \$71.64 against a \$71.59 Friday close, which came **within 0.9%** of CME's Sunday reopen which landed at \$72.30.



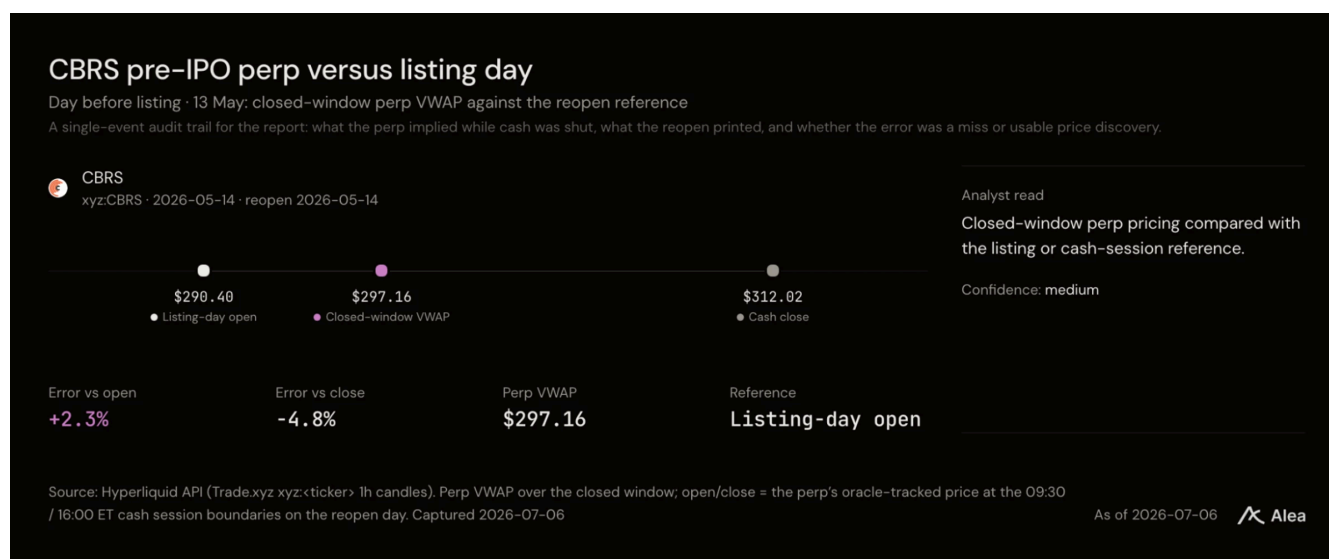
Micron reported their earnings after the 4PM close on 24 June, and unlike the Iran weekend a competing venue existed, since MU still trades post-market to 8PM and pre-market from 4AM. The dark window is the overnight session from 8PM to 4AM. What the perp offers here is leveraged, continuous access through that gap rather than sole price discovery.



The overnight VWAP printed \$1,202.71, **within 0.9%** of the \$1,213.80 close, and finished **3.3% below** the \$1,243.20 open after the stock extended its earnings pop into the morning session. The perp absorbed the reaction overnight and ran slightly cold, undershooting a rally rather than amplifying it.

Pre-IPO is the hardest to test because the perp prices a company with **no live market to anchor** against at all, only the last funding round which can serve as a soft price floor, and whatever the orderbook believes in the stretch before listing.

Cerebras (CBRS) was the first pre-IPO test on Hyperliquid which traded only 13 days before Cerebras' IPO on 14 May. Against a \$185 offer price the stock opened at \$290.40, up 57%, and kept climbing to a \$312.02 close. The pre-IPO perp managed to land **within 2.3%** of the opening print and **4.8%** of the close.



This was a successful case study of pre-IPO equity trading, allowing unmet demand to price a stock more accurately than the IPO underwriters.

SpaceX was the largest listing and the noisiest contract, overshooting past \$220 in May as retail enthusiasm ran hot before the S-1 filing revealed a \$135 IPO offer price, pushing the perp down. By the eve of the listing, the closed-window VWAP of \$167.81 came **within 2.7%** of the \$172.50 open on 12 June, and **4.0%** above the \$161.43 close.

SPCX pre-IPO perp versus listing day

Day before listing · 11 Jun: closed-window perp VWAP against the reopen reference

A single-event audit trail for the report: what the perp implied while cash was shut, what the reopen printed, and whether the error was a miss or usable price discovery.

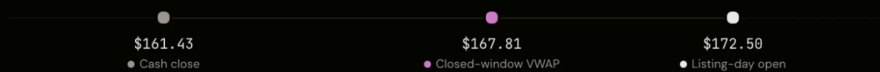
SPCX

xyz:SPCX · 2026-06-12 · reopen 2026-06-12

Analyst read

Closed-window perp pricing compared with the listing or cash-session reference.

Confidence: medium



Error vs open

-2.7%

Error vs close

+4.0%

Perp VWAP

\$167.81

Reference

Listing-day open

Source: Hyperliquid API (Trade.xyz xyz:<ticker> 1h candles). Perp VWAP over the closed window; open/close = the perp's oracle-tracked price at the 09:30

/ 16:00 ET cash session boundaries on the reopen day. Captured 2026-07-06

As of 2026-07-06

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Quantinuum's IPO was the first and only stress test which saw the pre-listing VWAP of \$99.38 land **within 5.6%** of the \$94.18 open. Then the stock collapsed through its own debut, sliding to a \$60.28 close which meant the perp's pricing was **65%** off. While the pre-IPO perp is capable of pricing the demand that shows up at the open, it cannot see the supply that emerges once real float starts trading.

QNT pre-IPO perp versus listing day

Day before listing · 3 Jun: closed-window perp VWAP against the reopen reference

A single-event audit trail for the report: what the perp implied while cash was shut, what the reopen printed, and whether the error was a miss or usable price discovery.

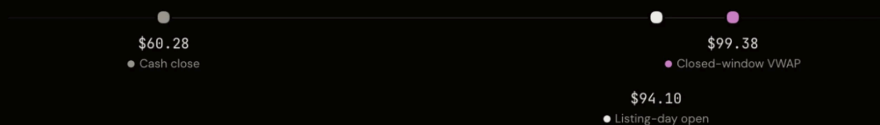
QNT

xyz:QNT · 2026-06-04 · reopen 2026-06-04

Analyst read

Predicted the open, then collapsed intraday on debut (open \$94 → close \$60).

Confidence: medium



Error vs open

+5.6%

Error vs close

+64.9%

Perp VWAP

\$99.38

Reference

Listing-day open

Source: Hyperliquid API (Trade.xyz xyz:<ticker> 1h candles). Perp VWAP over the closed window; open/close = the perp's oracle-tracked price at the 09:30

/ 16:00 ET cash session boundaries on the reopen day. Captured 2026-07-06

As of 2026-07-06

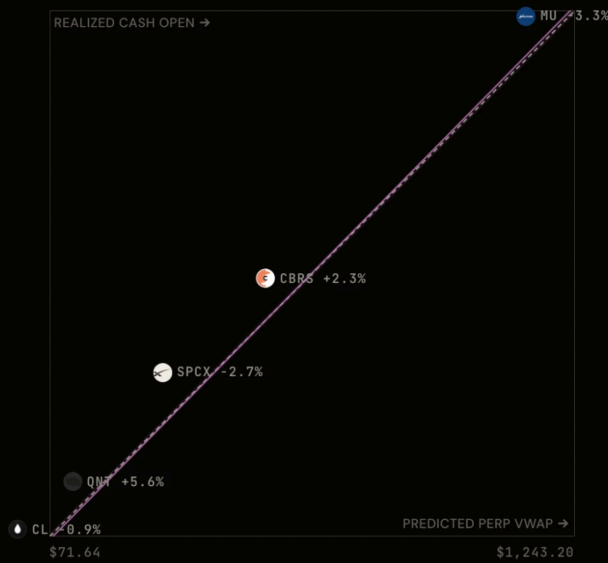
Alea

Across these 5 separate events, the **mean error is +0.2%**, meaning RWA perps have no systematic bias. This is important because the obvious failure mode for a retail-driven synthetic market is consistent overpricing as hype pushes every read above reality. The **median error** which shows the typical miss size ignoring direction, is **2.7%**. 4 of the 5 examples above landed within 5% of the open.

Did the closed-window perp land the cash open?

Perp VWAP over the shut window vs the price the cash session reopened at

Each point is one overnight or pre-IPO event: the perp's volume-weighted price while the cash market was shut (predicted) against the reopen price (realized), on log axes. The dashed 45° line is perfect calibration; the shaded band is $\pm 2\%$. The perp tracks the level across an $18\times$ price range, so the grade is the bias and hit-rate, not a price R^2 . Curated + cited; 5 events.



MEAN SIGNED BIAS

+0.2%

Across 5 events, vs the cash open. Near zero = unbiased.

MEDIAN ABSOLUTE ERROR

2.7%

LANDED THE OPEN WITHIN

±1%		20.0%
±2%		20.0%
±5%		80.0%

MULTIPLICATIVE FIT SLOPE

1.015 1.00 = no systematic level bias

Source: Hyperliquid API (Trade.xyz xyz:<ticker> 1h candles). Perp VWAP over the closed window; open/close = the perp's oracle-tracked price at the 09:30 / 16:00 ET cash session boundaries on the reopen day. Captured 2026-07-06 · Calibration = perp closed-window VWAP vs the reopen price; corridor is $\pm 2\%$. Small curated sample; bias and hit-rate are the grade, not a price R^2

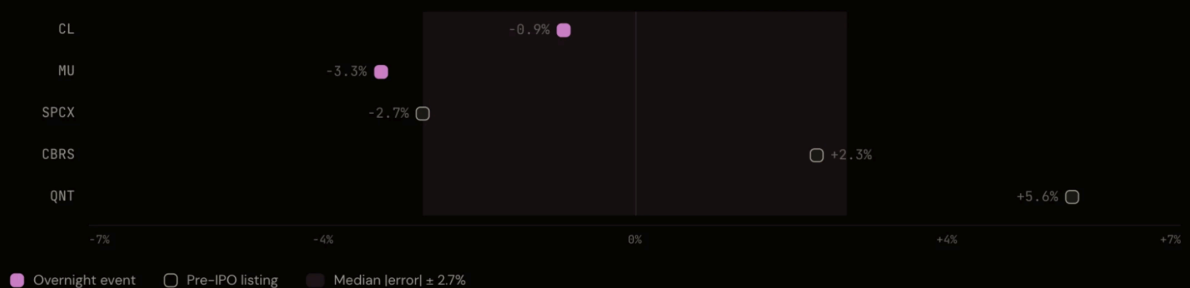
As of 2026-07-06 Alea

Viewing this as an error band makes the pattern easier to see. The overnight shocks (WTI, MU) are clustered tighter than pre-IPO listings, but they all fall within single digit percentages of the opening price.

Pricing accuracy, as a distribution

Signed error of the closed-window VWAP vs the cash open, across every event

0% means the perp priced the reopen exactly. The band spans the median absolute error; overnight events and pre-IPO listings are marked separately.



Source: Hyperliquid API (Trade.xyz xyz:<ticker> 1h candles). Perp VWAP over the closed window; open/close = the perp's oracle-tracked price at the 09:30 / 16:00 ET cash session boundaries on the reopen day. Captured 2026-07-06

As of 2026-07-06 Alea

Perp DEX and CEX RWA Perps

Perp DEXes first introduced the concept of RWA perpetuals, with CEXes following suit. Every venue has roughly the same architecture, issuing synthetic contracts that hold no underlying share., using funding rates to pull the mark price toward the oracle, and have deviation caps on weekends to stop thin books from large swings.

DEX vs CEX: who wins on what

The two RWA-perp models graded across the axes that decide where flow goes

A structured read, not a leaderboard: which model leads on volume, depth, breadth, pre-IPO, transparency and access. Curated synthesis — the volume shares are as reported for the RWA-perp month, the edges are the desk call.

DEX leads 4 CEX leads 2 Ties 2

On-chain venues win breadth, transparency and capital efficiency; CEXs still win raw volume and depth. The migration lives where those overlap.

Dimension	DEX · Hyperliquid HIP-3	CEX · Binance-led	Edge
Monthly volume Binance-led CEXs still clear roughly twice the on-chain RWA-perp notional.	28.9% · ~\$61B	55.7% · ~\$118B	CEX
Depth at size CEX liquidity absorbs size; thin on-chain weekend books gap on the 1%-per-update mark clamp.	Thinner; weekend gap-risk	Deeper books	CEX
Listing breadth On-chain builders list the long tail — single names, indices, commodities — a CEX will not.	135 HIP-3 markets, single-name long tail	Curated majors	DEX
Pre-IPO & private names HIP-3 is the on-chain pre-IPO venue after Ventuals wound down; CEXs rarely list them.	Real-price SPCX / QNT / CBRs	Minimal	DEX
Capital efficiency The on-chain book turns its open interest over faster — younger, more momentum-driven flow.	~1.4x vol/OI per day	Lower turnover	DEX
Oracle transparency HIP-3 marks are relayer / Stork / Chainlink feeds anyone can verify; a CEX mark is a black box.	On-chain oracle, auditable mark	Internal mark	DEX
Access & custody A real trade-off: on-chain sovereignty versus regulated custody and US access.	Self-custody, no-KYC, geo-blocked	KYC, regulated, broad reach	tie
Settlement & continuity Both run around the clock; neither closes for the cash session.	USDC, continuous on-chain	Exchange-internal, 24/7	tie

Source: Alea synthesis — RWA-perp market split (CoinDesk / Artemis dashboard capture, curated) with Hyperliquid HIP-3 live listing breadth and per-venue oracle / access / pre-IPO profiles. Volume shares as reported for the RWA-perp month; edges are the desk read, not a live feed

As of 2026-07-08

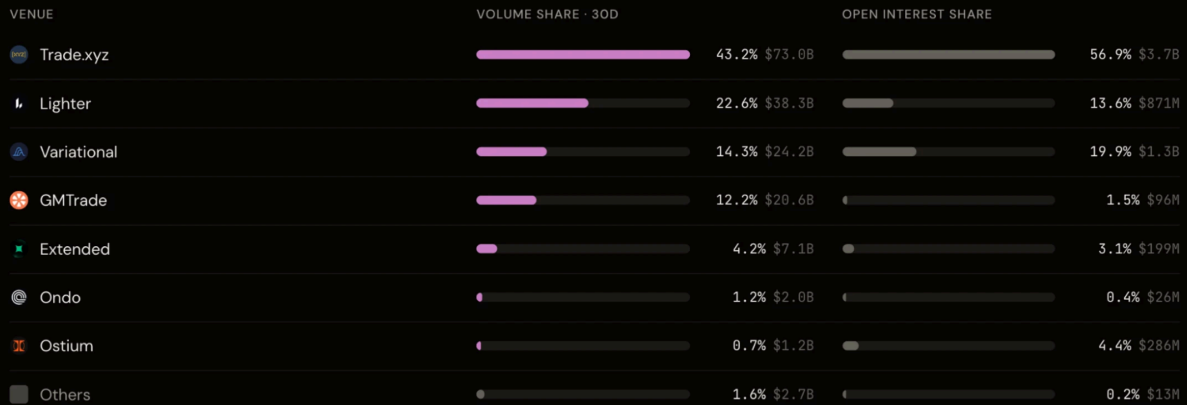


Hyperliquid's [HIP-3](#) builder-deployed perpetuals saw the first iteration of RWA perpetuals launched by TradeXYZ. A builder stakes 500k HYPE, sets the oracle and risk parameters, and lists a market in days compared to traditional exchanges who spend weeks on legal review and potential market maker negotiations for a single contract. **TradeXYZ holds 56.9%** of onchain RWA perps open interest with multiple listed US tickers, commodities, FX, and pre-IPO contracts at roughly \$3.1B.

Equity-perp volume and open interest by venue

Share of the RWA-perp market, trailing 30 days

The seven named venues individually; other RWA-native venues folded into Others. Volume is trailing-30-day notional; open interest is the latest snapshot.



Source: DefiLlama Pro derivatives, per-venue notional volume and open interest · Venue totals include each venue's full perp book; mixed crypto+RWA venues (Variational, Lighter, Extended) are not RWA-only

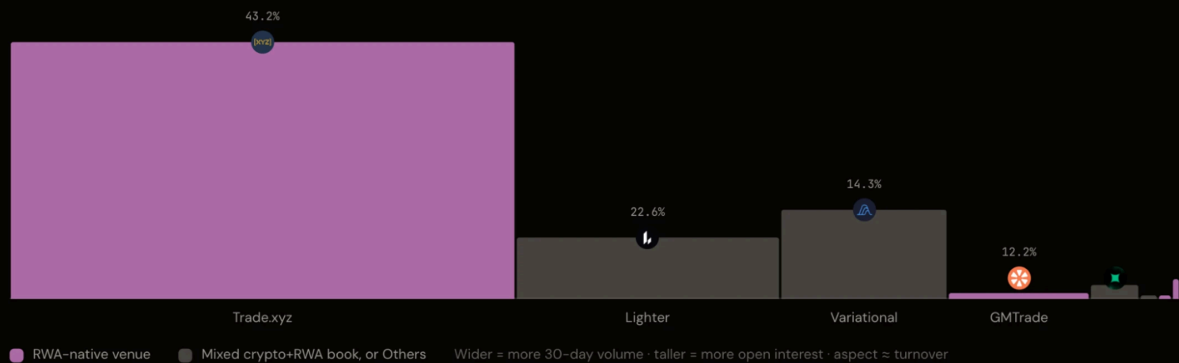
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Volume and open interest rank the platforms differently. TradeXYZ leads on both volume and open interest, which means capital trades there and stays on the book as held positions. Lighter and GMTrade show 22.6% and 12.2% of volume, resting on just 13.6% and 1.5% of open interest respectively. This shows **higher volume churning** activities rather than organic positioning.

The venue landscape, size against capital

Column width = 30-day volume share · height = open interest

A wide, short column trades a big flow on a thin book (high turnover); a tall, narrow one holds sticky positions. RWA-native venues in accent; mixed crypto+RWA books and Others muted, their DefiLlama totals overstate RWA-only activity.



Source: DefiLlama Pro derivatives, per-venue trailing-30-day notional volume and latest open interest · Width is volume share, height is open interest; mixed crypto+RWA venue totals overstate RWA-only activity

As of 2026-07-12 Alea

GMTrade, Variational, and Lighter compete on execution models rather than listing breadth, each pulling a smaller share of RWA flow. For example, Lighter streams Chainlink equity feeds while Variational prices through an RFQ that pulls the best bids across multiple platforms. How a venue sources a price when the underlying market is closed is exactly where these designs diverge.

Venue table							
Model, access, off-hours pricing, equity-RWA scope, live 30-day volume and OI							
The report table pairs curated venue metadata with DefiLlama live volume/OI where the venue is listed. Mixed crypto/RWA venues are labelled so totals are not over-read as equity-only flow.							
Venue	Model	Scope	Off-hours pricing	30d volume	OI	Differentiator / risk	
 Trade.xyz Self-custody, no KYC, geo-blocked (US/Ontario); settles USDC	HIP-3 builder	Equities, indices, commodities, pre-IPO US single names, SPX/QQQ style indices, commodities Pre-IPO: Real-price pre-IPO (SPCX/QNT/CBRS); the on-chain pre-IPO venue after Ventuals wound down	XYZ relayer marks ~every 3s; HIP-3 markPx clamped to 1% per update — weekend gap-risk in thin liquidity	\$73.0B	\$3.7B	Fastest HIP-3 RWA breadth and pre-IPO experimentation Builder-specific markets can be invisible on generic Hyperliquid front ends.	
 Lighter zk-rollup with self-custody escape hatch; trading-layer KYC/geo unconfirmed	DEX	Crypto-first perps, expanding into equities Selective equity perps Pre-IPO: None	Chainlink feed; 24/7 equity streams for off-hours (mechanism undisclosed)	\$38.3B	\$874M	Fast matching and broad perp distribution RWA-only volume cannot be isolated from venue total here.	
 Variational On-chain escrow per trade; Restricted-Persons ToS gate (not fully permissionless)	Hybrid	Mixed crypto and RWA derivatives Selective RWA expansion Pre-IPO: None (commodities-first)	RFQ mark = best pooled quote from CEX/DEX/TradFi dealers (Omni Liquidity Provider), not an index oracle	\$24.2B	\$1.3B	Institutional-style derivatives architecture DefiLlama venue totals include crypto flow.	
 GMTrade Self-custody, no KYC, Solana	DEX	RWA-native perps Stocks and commodities Pre-IPO: None	Oracle-priced GLV pools, not order-book depth; weekend mechanics undisclosed	\$20.6B	\$95M	Smaller RWA-native venue in the long tail Liquidity is thinner than Trade.xyz/Hyperliquid.	
 Extended Reachable from 6 EVM chains without bridging; custody/KYC undisclosed	Hybrid	Crypto plus selected RWA perps Selective equities Pre-IPO: None	Starknet order book; off-hours mechanics undisclosed	\$7.1B	\$199M	CEX-like UX with perp DEX settlement model Mixed book overstates RWA-only activity.	
 Ondo Non-US only; first to accept tokenized-stock holdings as perp collateral	DEX	Tokenized finance and RWA perps RWA-oriented coverage Pre-IPO: None (equities/ETFs/commodities)	New (Jul 2026); off-hours mechanics undisclosed	\$2.0B	\$26M	RWA brand distribution and tokenized-asset adjacency Coverage is early and may be episodic.	
 Ostium Non-custodial, no KYC, country-checker geo-gate; USDC on Arbitrum	DEX	Commodities, FX, indices, selected equities Macro-heavy RWA set Pre-IPO: None	Stork Network custom oracle acts as price-setter, tracking underlying spot directly	\$1.2B	\$286M	Macro/RWA specialist with explicit non-crypto positioning Venue data is not one-to-one comparable with HIP-3 per-market data.	
Source: DefiLlama Pro derivatives volume/OI, joined to curated venue profiles from the Equity Perps report audit						As of 2026-07-12 	

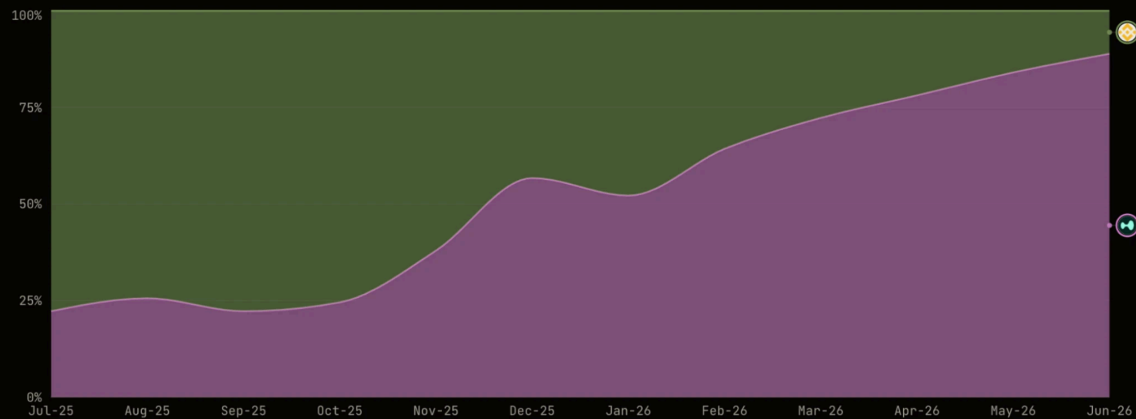
CEXs moved into RWA perps once the DEXes proved strong demand. CEXes in comparison have a much larger KYC'ed user base and fiat rails. RWA perps on DEXes continue to dominate in terms of volume, averaging ~\$24B weekly in June, with CEXes at ~\$3B weekly.

RWA perps: the DEX flipping

Share of monthly RWA-perp notional, DEX vs CEX

Permissionless listing pulled the long-tail and offshore flow on-chain: DEX RWA-perp volume climbed from roughly a fifth of the market to ~89% over the year, ~\$24B/week vs ~\$3B on CEXs by June. Curated monthly capture; complements the live point-in-time venue panels.

DEX · Hyperliquid-led **\$24.0B/wk 88.9%** CEX · Binance-led **\$3.0B/wk 11.1%** Crossover Dec-25 · as of Jun-26



Source: Artemis RWA Perps Sector Dashboard and DefiLlama Perps Dashboard; monthly capture · Curated monthly capture; DEX carries the Hyperliquid mark, CEX the Binance mark as venue-type representatives

As of 2026-07-09 Alea

If we look at the Vol/OI multiple, it has dropped significantly since July 2025, with the multiple averaging **8.36x for DEXes** and **5.11x for CEXes** showing healthy demand-driven usage as traders carry exposure **across sessions** rather than churning them intraday. One caveat for DEXes which could contribute to higher multiples is the speculation aspect from point farming.

How fast RWA perps turn over

Daily notional ÷ open interest, DEX vs CEX

A high multiple is fast, then-OI, speculative flow; a low one is position-heavy carry. Both sides fell from the Jul-25 highs toward the report's 8.36x (DEX) / 5.11x (CEX) window averages — usage maturing from farming toward carried exposure. Curated monthly capture.



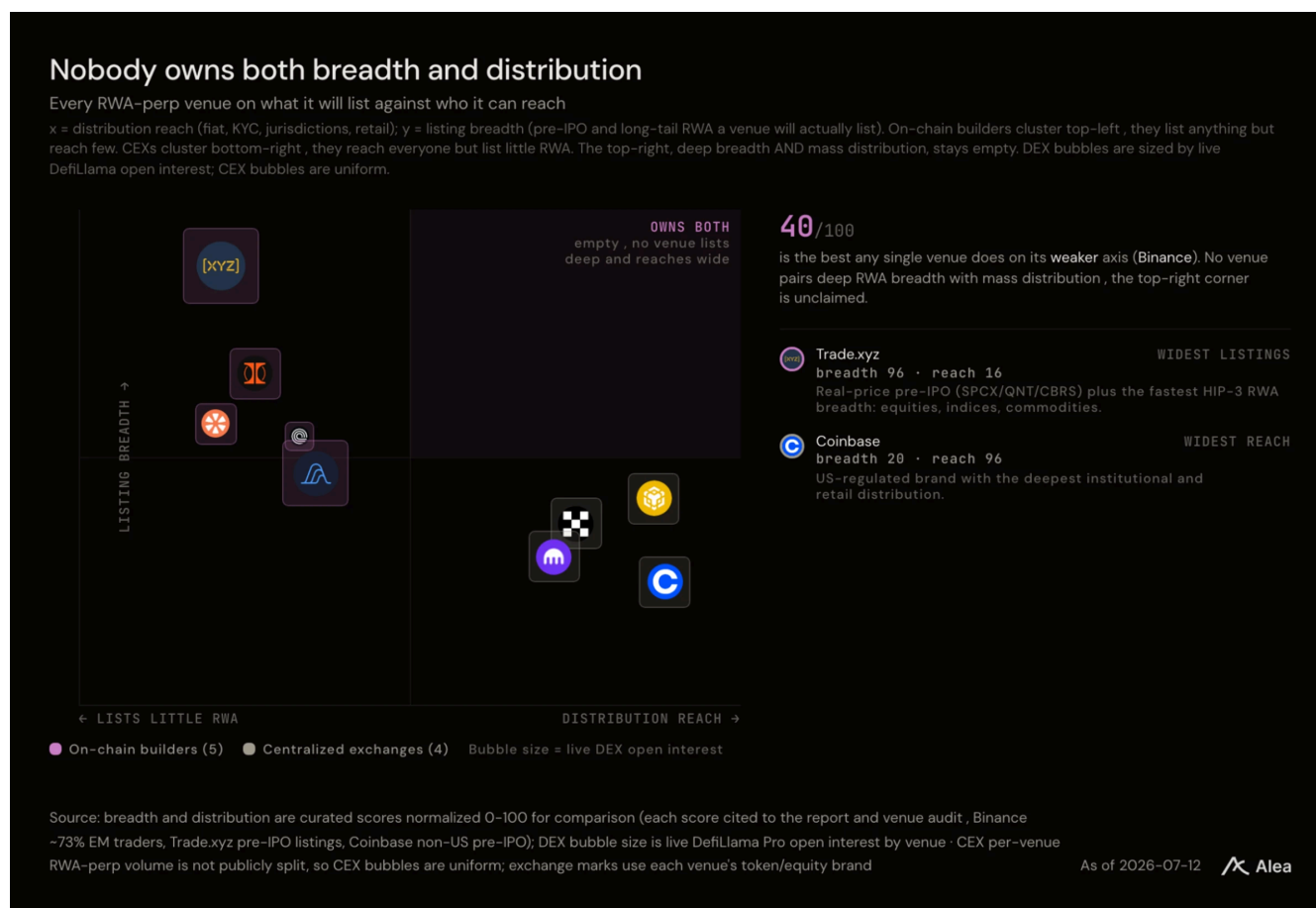
Source: Artemis RWA Perps Sector Dashboard and DefiLlama Perps Dashboard; monthly capture · Curated monthly capture; DEX carries the Hyperliquid mark, CEX the Binance mark as venue-type representatives

As of 2026-07-09 Alea

Binance sits at the retail end. They [recently reported that 73%](#) of its stock product traders come from **emerging markets**. User geography on perp DEXes is hard to verify, but Binance's number shows CEX distribution can put RWA exposure in front of traders in crypto's highest-adoption regions, where **access to U.S. brokerages is limited**.

Kraken sits at the compliant, institution-adjacent end. Its perps run through a [regulated derivatives venue](#) with full onboarding, which opens a door no DEX can, since a registered adviser or asset manager is barred by its own **compliance and custody rules** from trading against an anonymous onchain order book no matter how deep HIP-3 liquidity gets. OKX and Coinbase occupy the same regulated lane, with Coinbase adding pre-IPO perps for non-US users alongside its spot listings.

Both camps list pre-IPO perps, and they price them the same way, off a self-referential mark rather than a live underlying, because the company has no public share until it lists. Their mechanics might differ slightly. For instance, OKX [rebases the contract](#) when the S-1 discloses the share counts, Kraken derives its index from a moving average of the contract's own mark and transitions to the xStock market at listing.



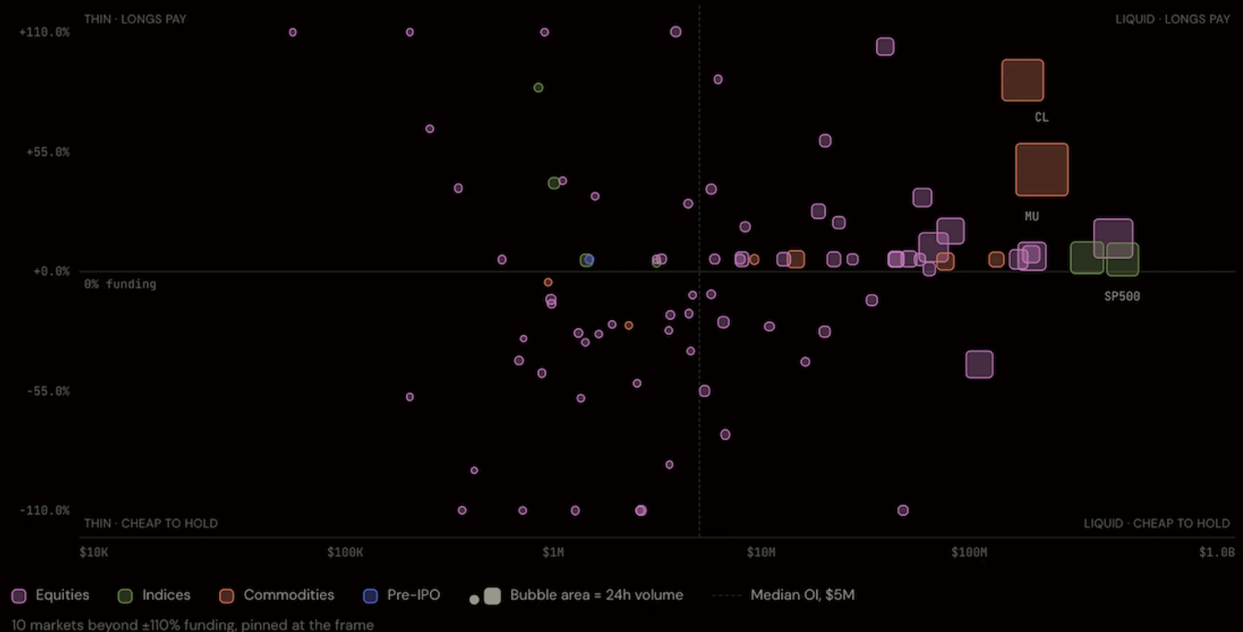
Perp DEXs hold the widest name coverage because permissionless listing reaches jurisdictions and long-tail tickers no licensed venue prioritizes, which keeps crypto-native and offshore traders there. CEXes house capital that is restricted from touching DEXes because a KYC venue is the only access point for mandated institutions, and even emerging market retail traders.

The tradable universe, mapped

Every on-chain equity, index, commodity and pre-IPO perp by liquidity and carry

Each bubble is one HIP-3 market: horizontal = open interest (the size it holds), vertical = annualized funding (what a long pays to hold it), area = 24h volume, color = asset class. Deep-and-cheap markets (lower-right) are the sweet spot for long exposure; thin-and-rich (upper-left) the most expensive to crowd into.

86 markets · median funding +5.5% APR · deepest SP500 \$544M OI



Source: Hyperliquid API, HIP-3 builder markets (open interest, annualized funding hourly $\times 24 \times 365$, and 24h volume) across every RWA and real-price pre-IPO market

As of 2026-07-12 Alea

Conclusion

Overnight order books and private secondaries were already full of traders who **could not reach the market they wanted**, and crypto rails gave them the **only venue open** when everything else was closed. How accurately equity perps price these events is the test of whether this is product-market fit or just a new place to gamble.

When the regulated venues finally run 23/5, the round-the-clock edge disappears, and what remains is the weekend, the leverage, and the names TradFi cannot list. **RWA perp open interest** will be the key data point to watch during the night sessions that quarter. If it holds, the demand drives the usage. If it flattens, access was doing the work all along, and the market is smaller than the volume made it look.

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